



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Density

1.4

STUDENT NAME



0315 0805775

PHYSICS WITH
MOIZ DAWOODI

1.4 Density

- 1 Define density as mass per unit volume; recall and use the equation

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\rho = \frac{m}{V}$$

- 2 Describe how to determine the density of a liquid, of a regularly shaped solid and of an irregularly shaped solid which sinks in a liquid (volume by displacement), including appropriate calculations

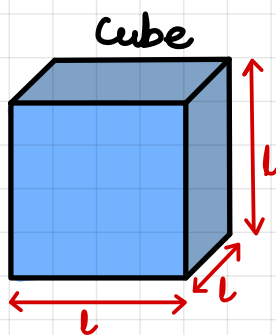
VOLUME.

"The space occupied by a body is called volume."

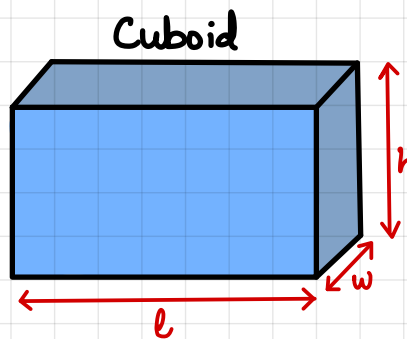
Its S.I unit is cubic meter (m^3).

VOLUME OF REGULARLY SHAPED BODIES:

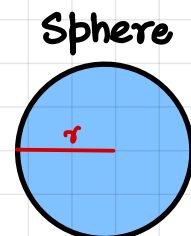
Volume of regularly shaped bodies can be found by formula.



$$V = l^3$$



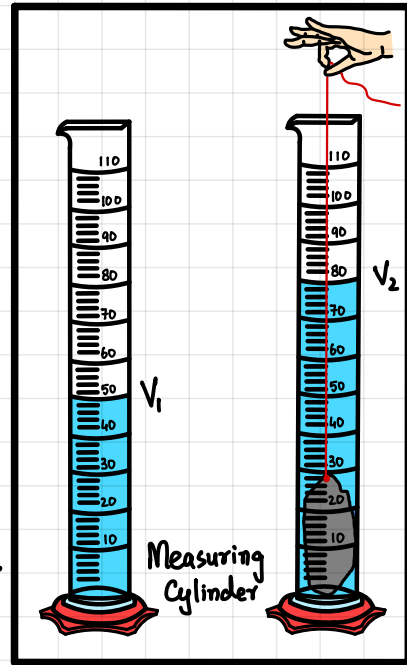
$$V = L \times w \times h$$



$$V = \frac{4}{3} \pi r^3$$

VOLUME OF IRREGULARLY SHAPED BODIES:

- Pour fixed amount of water in measuring cylinder & name the volume of water as ' V_1 '.
- Tie irregular stone with a thread & lower stone in measuring cylinder.
- Shake measuring cylinder gently to remove air bubbles.
- There will be new level of water. Mark this as ' V_2 '.
- To find volume of stone, subtract $V = V_2 - V_1$.



DENSITY:

"Density is the ratio of mass over volume."

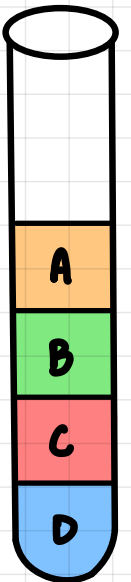
FORMULA:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$\rho = \frac{m}{V}$$

Its S.I unit is $\frac{\text{kg}}{\text{m}^3}$.

Liquid A is least dense while liquid D is most dense.



DENSITY OF IRREGULARLY SHAPED BODIES:

- Measure mass using electronic balance.
- Measure volume by either displacement method or eureka method.
- Divide mass by volume to find density.

DENSITY OF LIQUIDS:

- Measure mass of empty measuring cylinder using electronic balance.
- Measure mass of filled measuring cylinder using electronic balance.

The volume of liquid is v , measured by measuring cylinder

- Subtract both masses to find mass of liquid.
- Divide mass by volume to find density.



MATCH THE FOLLOWING



Density is the ratio of

nature of material doesn't change.

Density remains same if

mass over volume.

Solids have higher density

upwards.

Less dense thing tends to go

than liquids.